Mansi Phute

Mansi Phute (Adversarial ML + Explainable AI)

My research focuses on the **security** and **explainability** of language and vision foundation models. I work on developing explanations for machine learning models and analyzing them to identify vulnerabilities in existing ML systems, then find solutions to mitigate these issues. My work spans a wide range of application areas, including robust multi-object tracking in computer vision, developing defenses against attacks on large language models, and understanding large language models and the insights they can give us into human interactions.

I have collaborated with researchers, designers, and developers, at Intel Labs, Dassault Systems and Nanyang Technological University.

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CV PDF
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Google Scholar

Education

Fall 2022 — Present

M.S. in Computer Science

Georgia Institute of Technology, Atlanta, GA
Specialization: Machine Learning

Fall 2018 — Spring 2022

B.Tech. in Electronics and Telecommunication

Vishwakarma institute of Technology,
Honors: Artificial Intelligence and Data Analytics
🏆 Highest GPA
🏆 Top 3 Best Graduating Students

Academic Research Experience

Summer 2022 — Present

Georgia Institute of Technology, Atlanta, GA

Graduate Research Assistant, School of Computational Science and Engineering
Advisor: Duen Horng (Polo) Chau
Member of the Polo Club of Data Science where we bridge and innovate at the intersection of data mining and human-computer interaction to synthesize scalable,

interactive, and interpretable tools that amplify human's ability to understand and interact with big data. Developed defences against adversarial attacks in Language and Vision domain.

Spring 2023

Georgia Institute of Technology, Atlanta, GA

Graduate Teaching Assistant, School of Computational Science and Engineering

Mentor: Duen Horng (Polo) Chau

Fall 2021 — Spring 2022

Nanyang Technological University, Singapore

Undergraduate Research Assistant, Cyber Security Research Centre at NTU (CYSREN)

Mentor: Thambipillai Srikanthan

Increasing python application security by analyzing libraries used. Developed dynamic dependency graph to trace vulnerabilities.

Fall 2021 — Spring 2022

Nanyang Technological University, Singapore

Undergraduate Research Assistant, Cyber Security Research Centre at NTU (CYSREN)

Mentor: Thambipillai Srikanthan

Automated human resource planning and forecasting by combining business intelligence of NHS, UK with data analytics to properly shift the HR planning from manual to automated

Spring 2021

Vishwakarma Institute of Technology, India

Undergraduate Research Assistant, Associated with Dassault Systems

Mentor: Jyoti Madake

Developed AI based solutions for agricultural problems faced in India by usimg hyperspectral imaging to predict soil fertility in the land

Industry Experience

Summer 2019

Tech Mahindra Ltd, Pune, India

Intern, Web Development,

Mentor: Rahul Bedmutha

Developed AI based solutions for agricultural problems faced in India by usimg hyperspectral imaging to predict soil fertility in the land

Honors and Awards

2022

Best Scholar in ECE

Merit-based award for the ECE student with the highest undergraduate GPA in the entire department

2022

Best Student in ECE

One of the 3 students chosen as the best student in ECE department based on overall performance throughout undergrad

2015 Times NIE Student of the Year

Selected by "Times of India" as the best student in school based on overall acedamic and extracurricular performance.

Publications

Selected: Latest & Greatest

LLM Self Defense: By Self Examination, LLMs Know They Are Being Tricked

Mansi Phute, Alec Helbling, Matthew Hull, ShengYun Peng, Sebastian Szyller, Cory Cornelius, Duen Horng (Polo) Chau

Under review. .

Project PDF BibTeX

All Publications

Conference

C3 **Robust Principles: Architectural Design Principles for Adversarially Robust CNNs**

ShengYun Peng, Weilin Xu, Cory Cornelius, Matthew Hull, Kevin Li, Rahul Duggal, Mansi Phute, Duen Horng (Polo) Chau, Jason Martin

British Machine Vision Conference (BMVC). 2023.

Project PDF Code BibTeX **#1 on RobustBench CIFAR-10 leaderboard , Best Poster**

C2 **A Survey on Machine Learning in Lithographys**

Mansi Phute, Aditi Sahastrabudhe, Sameer Pimparkhede, Shubham Potphode, Kshitij Rengade, Swati Shilaskar

IEEE International Conference on Artificial Intelligence and Machine Vision. 2021.

Project BibTeX

C1 **Precision Farming Based Soil Fertility Assessment Techniques**

Mansi Phute, Jyoti Madake, Sripad Bhatlawande

IEEE International Conference on Advances in Computing, Communication, Embedded and Secure Systems. 2021.

Project BibTeX

Poster

P1 **LLM Self Defense: By Self Examination, LLMs Know They Are Being Tricked**
Mansi Phute, Alec Helbling, Matthew Hull, ShengYun Peng, Sebastian Szyller, Cory Cornelius, Duen Horng (Polo) Chau
Under review. .
Project PDF BibTeX

Miscellaneous

M1 **Semantic Shift in Online Communities: Unmasking Information Flow through Linguistic Fingerprints**
Julia Kruk*, Mansi Phute*, Amit Bhatacharjeee, Sanchita Porwal
Under Review. .
→ More coming soon! Project *Authors contributed equally

Press

August 2023 "GRE Success Stories: How Test Takers Scored Above the 90th Percentile," Jamboree

May 2020 "Team Eklavya- E&TC; students team Designs Autonomous sanitisation robot," Vishwakarma Institute of Technology

Teaching

Spring 2023 **Graduate Teaching Assistant**
Georgia Institute of Technology, Atlanta, GA
Data and Visual Analytics (CSE 6242 / CX 4242), Instructor: Duen Horng (Polo) Chau
I was a Teaching Assistant (TA) at Georgia Tech for the class Data and Visual Analytics where I worked with a team of 30 TAs to enable learning in a class of more than 1200 students. I was a part of designing homework and mentoring students in their course work and project work.

Grants and Funding

2023 — 2024 **Guaranteeing AI Robustness against Deception**
DARPA: Defense Advance Research Projects Agency
Co-PIs:
Funded

Service

Reviewer
NeurIPS Workshop on Socially Responsible Language Modelling (**NeurIPS SoLaR**) 2023

References

Dr. Polo Chau, Associate Professor
School of Computational Science and Engineering
Georgia Institute of Technology
cc.gatech.edu/~dchau/

Dr. Thambipillai Srikanthan, Professor
School of Computer Science and Engineering
Nanyang Institute of Technology
www.ntu.edu.sg/scse/about-us/past-chairs/prof-thambipillai-srikanthan

Contact

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